

Training Response and VO2max

PPARD, HFE, BDNF, resting heart rate and physical activity scores, and VO2max loci load,
exercise heart rate and recovery

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Short and honest genetic basis. VO2max and training response are very trainable, but its genetics are poorly mapped: few loci, small effects and almost no validated score. This report tells what is read and what is not read.

Favourable. The heart rate at rest is typical to low (22.7, 58.1 and 4.4 percentiles; loci load in 42.0). The HFE H63D heterozygous is associated with slightly higher VO2max in endurance athletes. The load of VO2max loci / cardiorespiratory fitness is in percentile 80.6 (low loci, but in favor).

Unfavourable. The genetic slope for physical activity is low: scores in the percentiles 14.2 (measured by accelerometer) and 7.4 (self-reported); load of loci from moderate to vigorous activity in 29.7. This describes the behavior (spontaneous desire to move), not the ability. The load of heart rate recovery after effort is in percentile 15.6 (genetically slower recovery; it is very trainable). PPARG rs2267668 is heterozygous (some evidence of lower aerobic and metabolic response to training) and BDNF Val66Met is heterozygous (less activity dependent BDNF, that is, exercise counts even more for the brain).

What does it mean? There is no genetic sign of "non-responder", and the loci load of VO2max is favourable. The heart has a genetically favourable resting frequency and the genes of the aerobic motor are well (PPARGC1A and AMPD1 favourable; see the Sports Profile report and §27). The weak point is the spontaneous motivation for movement: physical activity must be structured and marked in the agenda, and not left to impulse. The individual response to training is measured (VO2max, heart rate), not by genes.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Keyvariants** 3 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 3 of the chip, 0 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 1 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 5 PGS Catalog scores calculated locally with 3.822.384 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MAIS PROVÁVEL/ALTO; ≤ 20 : MENOS PROVÁVEL/BAIXO; between 20 and 80: TÍPICO.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 270 loci, 252 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

At rest heart rate, a higher percentile = higher frequency (worst). In physical activity, a higher percentile = more genetic inclination to move (better).

Heart rate

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Resting heart rate	PGS002193 (Privé 2022, LDpred2, 858 thousand variants)	842.373 858.487 (weighted coverage 97%)	22,7	TIPIC	typical
Resting heart rate	PGS003472 (Zhang 2022, LDpred2, 777 thousand variants)	762.237 776.860 (weighted coverage 97%)	58,1	TIPIC	typical
Resting heart rate	PGS000300 (Xie 2020, 80 variants)	77 80 (weighted coverage 95%)	4,4	BAIXA	favourable

Physical activity

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Physical activity (measured by accelerometer)	PGS002255 (Kujala 2020, LDpred, 1.14 million variants)	1.107.375 1.140.182 (weighted coverage 96%)	14,2	BAIXA	unfavourable
Physical activity (self-reported)	PGS002254 (Kujala 2020, LDpred, 1.14 million variants)	1.110.322 1.142.416 (weighted coverage 96%)	7,4	BAIXA	unfavourable

4. Key variants by topic

Few variants with evidence of effect on training response; all have a small effect.

Aerobic response to training

The response to aerobic training (upper VO2max) varies greatly among people (intervention studies such as HERITAGE). The known genes explain only a small part.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PPARD	rs2267668		G/A	chip	G (1 copy/s)	unfavourable (light)	The G allele is associated with less improvement of aerobic capacity and insulin sensitivity with resistance training (intervention studies)	Weak

Oxygen transport

Oxygen transport depends on hemoglobin and available iron (HFE). The excess iron is harmful, so ferritin should be below 100.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
HFE	rs1799945	H63D	C/G	chip	G (1 copy/s)	favourable	H63D: the G allele slightly increases the available iron; associated with higher VO2max in endurance athletes (ferritin to maintain < 100; see §33)	Weak

Brain and exercise

Aerobic exercise is the strongest natural stimulus of BDNF (cerebral plasticity).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
BDNF	rs6265	V66M	C/T	chip	T (1 copy/s)	unfavourable (light)	Val66Met: T allele (Met) reduces activity-dependent BDNF release; aerobic exercise compensates part of this effect (cognitive detail in §34)	Moderate

5. GWAS loci burden

Sum of all loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of VO2max, heart rate (rest, exercise and recovery) and physical activity, with alleles oriented towards higher value, compared to the reference population. VO2max loci are few (tenenas).

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
VO2max / cardiorespiratory fitness	38 (35)	36 de 70	35	80,6	favourable
Resting heart rate	93 (89)	91 de 178	88	42,0	typical
Heart rate rise in exercise	29 (27)	21 de 54	27	16,9	without judgment
Heart rate recovery	57 (54)	55 de 108	53	15,6	unfavourable
Moderate to vigorous physical activity	53 (47)	47 de 94	47	29,7	typical

6. Favourable

- **Heart rate at typical to low rest:** percentiles 22.7, 58.1 and 4.4 (80 confirmed effect variants). A low resting frequency is associated with more cardiovascular longevity. **Moderate.**
- **HFE H63D C/G:** associated with slightly higher VO2max (more available iron); maintaining ferritin < 100. **Weak.**
- **VO2max / cardiorespiratory fitness:** loci load in percentile 80.6 (38 loci). **Weak moderated.**

7. Unfavourable

- **Genetic inclination for low physical activity:** percentiles 14.2 (accelerometer) and 7.4 (self-reported); loci load in 29.7. Moderate for behavior; no effect on capacity.
- **Heart rate recovery after exercise:** loci load in percentile 15.6 (57 loci). Slow recovery is associated with less fitness, but greatly improves with aerobic training. **Weak.**
- **PPARD rs2267668 A/G:** 1 allele associated with lower improvement of VO2max and insulin sensitivity with training (small studies). **Weak.**
- **BDNF Val66Met C/T:** less BDNF released with the activity; aerobic exercise is especially important for brain plasticity (§34). **Moderate.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Vascular Aging	Typical to low resting heart rate	Favourable (light)
Oxidative Phosphorylation Decline	PPARD rs2267668 heterozygous (lower mitochondrial response to training in some studies)	Unfavourable (mild)
Metabolic Homeostasis Decline	Little genetic inclination to physical activity: risk of sedentary lifestyle if not structured	Unfavourable
Intercellular Communication	BDNF Val66Met: less activity-dependent neurotrophic signal	Unfavourable (mild)
Fragility	Cardiorespiratory fitness is the major modifiable predictor of mortality; favourable VO2max loci, recovery of unfavourable heart rate	Mixed

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Structured and marked physical activity in the agenda** (the genetics of spontaneous motivation is low): fixed sessions, a partner or coach, and the objective of $\geq 8\,000$ – $10\,000$ steps/day measured per clock. *Suggestion (not applied).*
2. **Zone 2: 150–180 min/week**, plus 1–2 intervals sessions (e.g. 4×4 min to 85–95% of the maximum frequency). It is the most effective training in VO2max. *Suggestion (not applied).*
3. **Measure VO2max** (exercise test with base gas analysis) and 1×/year, to know the individual response to training, which the genes do not say. *Suggestion (not applied).*
4. **Follow heart rate at rest and heart rate variability** with the clock, as markers of adaptation and recovery. *Suggestion (not applied).*
5. **Iron:** without supplements; ferritin < 100 (H63D). If performance falls for no reason, see hemoglobin and ferritin. *Suggestion (not applied).*
6. **Aerobic exercise also by the brain** (BDNF Val66Met): preferably in the morning, consistent with caffeine ≤ 200 mg only in the morning. *Suggestion (not applied).*
7. **Recovery:** sufficient sleep and melatonin only at the dose of the plane (0.3–1 mg). Omega-3 3.2–4 g/day (plan). *Suggestion (not applied).*

10. Limitations

- The response to training (VO2max, strength) is ~50% hereditary in family studies, but the genes identified explain a minimal fraction. There are no validated polygenic scores of VO2max or training response.
- Physical activity scores measure behavior and are influenced by psychological traits and body composition.
- The HFE H63D is palindromic (C/G); the heterozygous is read without ambiguity.
- The loci load of VO2max is based on a few tens of loci; it is descriptive.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the Imposed VCF; in the palindromics the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog (5 scores: resting heart rate and physical activity). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 510^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** reference literature (Bouchard 2011 and Williams 2017 for training response; Stefan 2007 for PPARD; Eriksson 2018 for VO2max and heart rate; Kujala 2020 for physical activity Egan 2003 for BDNF), GWAS Catalog and ClinVar. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Aerobic response to training	PPARD	rs2267668	G/A	chip	G	1	strand +	1 copy of risk allele
Oxygen transport	HFE	rs1799945	C/G	chip	G	1	strand + (palindromic; assumed +)	1 copy of the protective allele
Brain and exercise	BDNF	rs6265	C/T	chip	T	1	strand +	1 copy of risk allele